HEDERA HACKATHON 2026

EL JOKER

5ALIHA 3LINA.

The control layer for AI agents that spend money. One AI proposes, four strict agents verify, decide and pay. Above the limit, only a human co-signature on Hedera opens the vault.

TEAM
CHAIN619

MEMBERS
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THE QUESTION

WHAT HAPPENS WHEN AI CAN SPEND MONEY?



WHO IS IN CONTROL?

Who stops it when it is wrong, or when someone tricks it?

And how can anyone prove, later, what the AI actually decided?



THE PROBLEM

THREE WAYS AGENTS FAIL

01



HALLUCINATE

Invents a supplier, a price or an approval, and acts on it.

02



OVERSPEND

Optimizes the task, ignores the budget, or pays the same invoice twice.

03



HIDE THE TRAIL

Acts first, explains later. Its logs can be edited after the fact.

Today you choose: autonomy or control. We remove that trade-off, for any organization that lets an agent spend.

WHERE AGENTS SPEND FIRST

ONE ENGINE. ANY ORGANIZATION.

RELIEF NGO · EMERGENCY

500 people in Gabès need water and medicine: **15,000 TND**, 48 hours.

Policy 2.1.0 · autonomous limit 15,000 TND

COMPANY · PROCUREMENT

12 laptops and 12 monitors for a Tunis team: **45,000 TND**, 7 days.

Policy 1.1.0 · autonomous limit 6,000 TND

Same agents, same proofs. Only the policy changes. We start with the hardest case: an emergency.



THE SOLUTION

AI PROPOSES. RULES DECIDE. HEDERA PROVES.

Autonomy for the routine. Hard limits on the money. Proof for everyone.

PROPOSE

Lemhaf · AI

Claude ranks the supplier offers.

VERIFY

Nasnes

Checks registry, evidence and the AI's claims.

DECIDE

3ases · no LLM

Fixed rules per organization: approve, block or escalate.

CONTROL

A human

Above the limit, co-signs on Hedera to release the money.

PAY & PROVE

Kanawita · Hedera

Pays only if authorized. Every step on HCS.

MEET THE CREW

FIVE AGENTS. ONE AI. ZERO CONFUSION.



LEM3ALEM	LEMHAF	NASNES	3ASES	KANAWITA
Coordinator	Research	Verifier	Guardian	Payment
Plans	The AI	Rules	No LLM	Rules + Hedera

HOW A MISSION FLOWS

REQUEST → PROOF

Every handoff is visible. Every decision leaves a receipt on Hedera.

REQUEST	PLAN	RESEARCH	VERIFY	GUARD	PAY	PROOF
The operator types the need	**Lem3alem** Valid mission, no guessed facts	**Lemhaf** Claude ranks the offers	**Nasnes** 10 independent checks	**3ases** Approve, block or human	**Kanawita** On testnet, exactly once	**Hedera** Every step on HCS

HUMAN CO-SIGNATURE

Above the limit: locked in a 2-of-2 Hedera vault until a human signs.

LIVE DEMO / APPROVED

15,000 TND BUDGET. MISSION CLEARED.

- Claude ranks 3 offers: **Oasis, 12,450 TND**
- Nasnes 10/10 checks · 3ases 12/12 checks, risk LOW
- Kanawita pays **12,450 EJTND**, our test dinar on Hedera
- 6 decisions anchored on Hedera, confirmed by the Mirror Node



Real run on Hedera testnet with Claude Opus 5.5 (first run, before the switch to dinars)

LIVE DEMO / BLOCKED

BLOCKED

FOUR WAYS TO SAY NO. MONEY NEVER MOVES.

B · BUDGET

300 TND BUDGET, 12,450 TND OFFER

3ases blocks. Kanawita never runs. The block itself is proven on Hedera.

D · UNVERIFIED SUPPLIER

THE AI PICKS THE CHEAP BROKER

Not in the registry, no license, prices half the market. Nasnes rejects, 3ases blocks.

C · DUPLICATE PAYMENT

SAME PAYMENT, SENT AGAIN

Rejected with HTTP 409: idempotency key plus a database constraint. Still one payment.

G · RECYCLED INVOICE

VERIFIED SUPPLIER, USED INVOICE

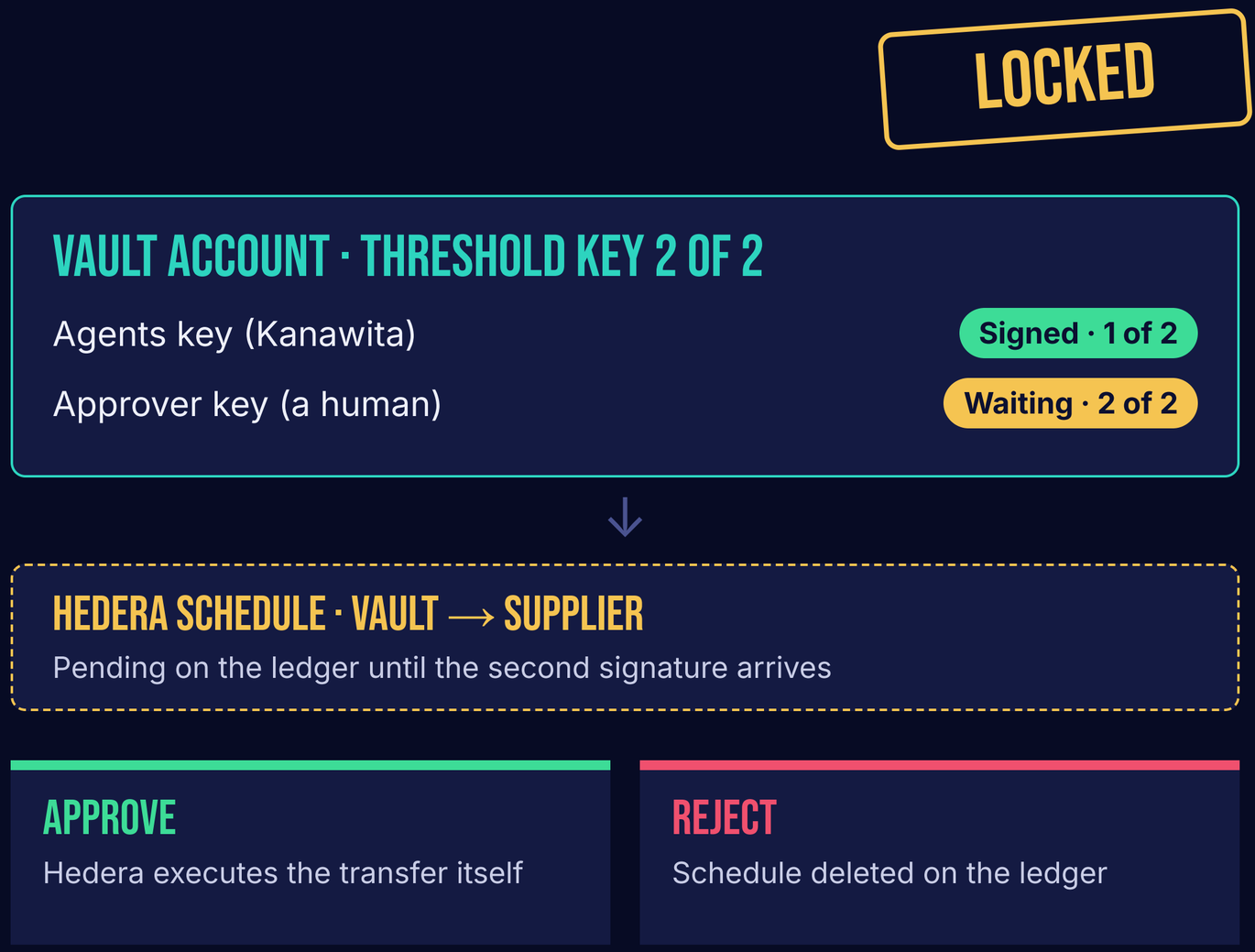
Its invoice number was already paid in an earlier mission. Rejected, blocked.

LIVE DEMO / HUMAN CO-SIGNATURE

ABOVE THE LIMIT, HEDERA HOLDS THE MONEY.

- ACME: **35,460 TND** of laptops, over its 6,000 TND limit
- Kanawita can only **lock** it: a Hedera schedule, 1 of 2 signatures
- Operator + reason + PIN → the approver signs → Hedera pays
- Reject → the schedule is deleted. Nothing moves

The agents' key alone can never move the vault. The approval is bound to the exact facts reviewed.



WHY 3ASES IS NOT AI

NOTHING TO TALK INTO A YES.

No LLM, no interpretation, no persuasion. Same facts + same policy version = same decision, every time.

IF OFFER ≤ BUDGET
AND SUPPLIER VERIFIED · REGION ALLOWED
AND NASNES = VERIFIED · INVOICE UNUSED
AND NO PRIOR PAYMENT · PRICE IN RANGE
THEN APPROVE · OVER THE LIMIT OR RISK HIGH ⟶ HUMAN
ONE FAILED RULE = BLOCK

13 rules · one versioned policy per organization (NGO 2.1.0, ACME 1.1.0) · 20 unit tests



HOW HEDERA IS USED

FOUR HEDERA SERVICES



No smart contract: Hedera enforces the human signature, carries the money and lets anyone check.

CONSENSUS SERVICE

Every decision fingerprinted (SHA-256), chained to the last one and published. Topic 0.0.11007004.

SCHEDULED TX + KEYS

The 2-of-2 vault: above the limit, the human co-signature is enforced by the ledger itself.

TOKEN SERVICE

EJTND, our test dinar: the full amount is paid in dinars, fixed supply, no admin key.

MIRROR NODE

Anyone re-checks everything, even from a browser, without trusting our server.

Only hashes on-chain: no names, prices or documents.

6 to 8 proofs per mission, refusals included.

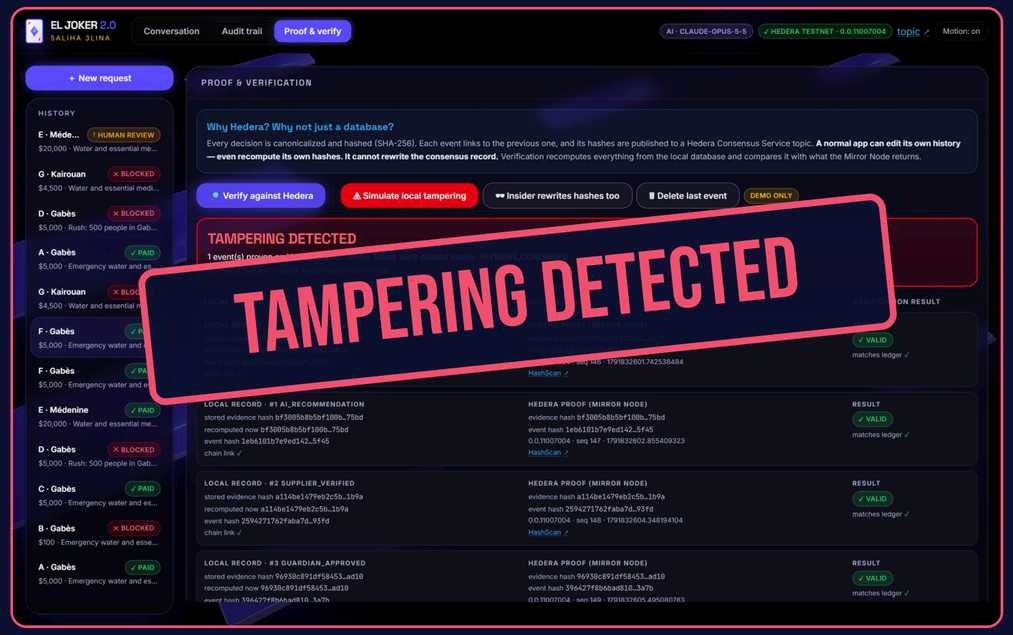
Native services: readable rules, low fixed fees.

WHY HEDERA, NOT A DATABASE

A DATABASE CAN REWRITE ITS PAST. HEDERA CAN'T.

Insider attack (demo)	Local check	With Hedera
Edit the approved price	Caught	Caught
Edit it and recompute every hash	Missed	Caught
Delete the payment record	Missed	Caught

Verified live on testnet. And you don't have to trust us: our **public verifier** recomputes every hash in your browser, reading Hedera directly.



Local record vs Hedera proof vs verification result

BUILT, TESTED, VERIFIED LIVE

FROM WORKING DEMO TO REAL DEPLOYMENT



8/8
scenarios passed live on Hedera testnet

115
automated tests, incl. 7 in a real browser

4
Hedera services: HCS, schedules, HTS, Mirror Node

2
organizations on one engine: an NGO and a company

NOW
PILOT-READY
Real suppliers, a licensed partner for the dinar token

NEXT
KEYS IN WALLETS
Approvers sign from their own wallet, real identity

THEN
TWO PILOTS
One NGO and one company in Tunisia

SCALE
POLICY LIBRARY
Policies per sector, shared supplier registry

Who pays: any organization that lets agents spend — SaaS per organization + per proof, NGOs first, then companies (our hypothesis).
Honest limits today: fictional suppliers, a test dinar token, a server-held approver key, a demo PIN.

AUTONOMOUS, BUT NEVER UNCONTROLLED.

Agents that act. Rules that cannot be argued with. Proof that anyone can check.

EL JOKER — 5ALIHA 3LINA

THANK YOU

github.com/rihamghoudi-bit/el-joker-2 · hashscan.io/testnet/topic/0.0.11007004 · verify any mission: /verify

